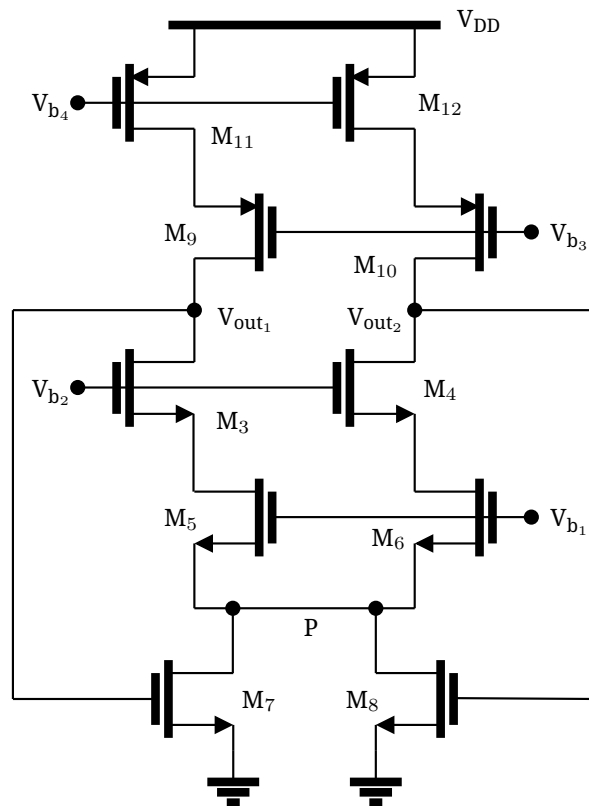


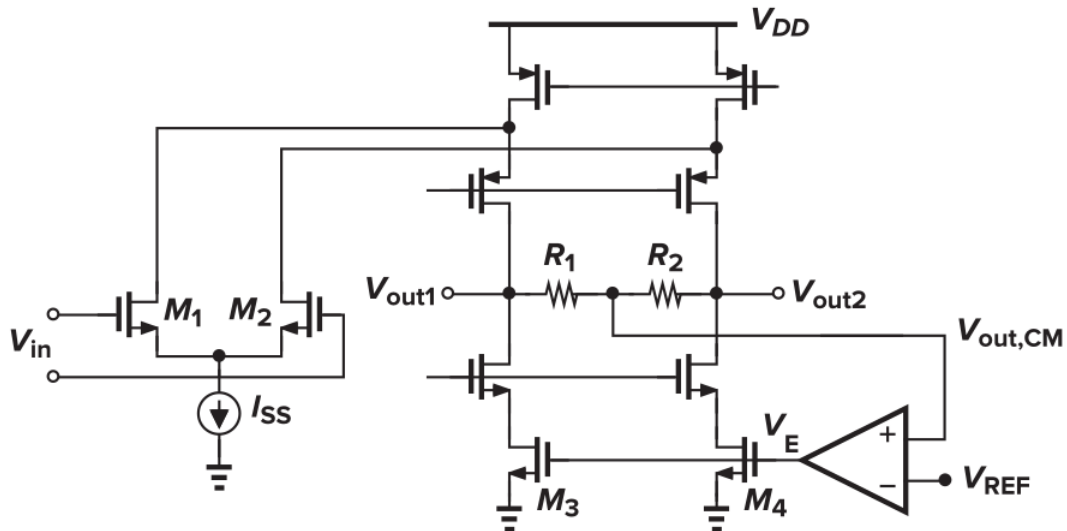
1. In the circuit of Figure below, each branch is biased at a current of 0.5 mA.

- Find the dimensions of M_7 and M_8 such that the output Common Mode level is equal to 1.5 V and $V_P=100$ mV.
- If all transistors have same size, what is the typical value of V_{b1} ?
- If output swing is decided by MOSFETs shown in diagram, what is the maximum differential output voltage swing?

Given: $V_{DD}=3$ V, $\mu_n C_{ox}=135 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox}=40 \mu\text{A}/\text{V}^2$, $V_{th,n}=0.7$ V, $|V_{th,p}|=0.8$ V, $\lambda_n=0.1 \text{ V}^{-1}$ at $L=0.5 \mu\text{m}$, $\lambda_p=0.2 \text{ V}^{-1}$ at $L=0.5 \mu\text{m}$.



2. Consider the CMFB network in Figure below. The amplifier sensing $V_{out,CM}$ is to be implemented as a differential pair with active current mirror load.
- (a) Should the input pair of the amplifier use PMOS devices or NMOS devices?
- (b) Calculate the loop gain for the CMFB network.



3. For the following circuit, $I_1 = 50 \mu\text{A}$, $I_2 = 200 \mu\text{A}$, $\left(\frac{W}{L}\right) = \frac{50\mu\text{m}}{1\mu\text{m}}$ for all MOSFETS. $R = 10 \text{ M}\Omega$. What is the differential gain and common mode gain of main amplifier without common mode feedback? What is the optimum value of $V_{O,CM}$? With common mode feedback circuit, if common mode output voltage optimum value is allowed to vary by $\pm 1\%$, then what is the expected value of common mode gain? (write this value till 3 decimal places). What is CMRR without common mode feedback? What is common mode loop gain? What is common mode gain with feedback (achieved with circuit)? What is CMRR with common mode feedback (achieved with circuit)?

Given: $V_{DD} = 1.8 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 50 \mu\text{A}/\text{V}^2$, $\lambda_p = 0.2 \text{ V}^{-1}$, $\lambda_n = 0.1 \text{ V}^{-1}$, $V_{th,n} = 0.4 \text{ V}$, $|V_{th,p}| = 0.5 \text{ V}$. Neglect body effect.

